

Collaboration-Based Learning Environments using Augmented Reality

by
Sebastian Gil Parga

Submitted to the Department of Creative Technologies
in partial fulfillment of the requirements for the degree of
DOCTOR OF PHILOSOPHY IN CREATIVE TECHNOLOGIES
at the
AUCKLAND UNIVERSITY OF TECHNOLOGY
November 2025

© 2025 Sebastian Gil Parga. All rights reserved.

Authored by: Sebastian Gil Parga
School of Future Environments, Department of Creative Technologies
November 30, 2025

Certified by: Stefan Marks
Associate Professor, Thesis Supervisor

Certified by: Jairo Gutierrez
Professor, Thesis Supervisor

Collaboration-Based Learning Environments using Augmented Reality

by

Sebastian Gil Parga

Submitted to the Department of Creative Technologies
on November 30, 2025 in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY IN CREATIVE TECHNOLOGIES

ABSTRACT

The developments of the “kinetic theory” of gases made within the last ten years have enabled it to account satisfactorily for many of the laws of gases. The mathematical deductions of Clausius, Maxwell and others, based upon the hypothesis of a gas composed of molecules acting upon each other at impact like perfectly elastic spheres, have furnished expressions for the laws of its elasticity, viscosity, conductivity for heat, diffusive power and other properties. For some of these laws we have experimental data of value in testing the validity of these deductions and assumptions. Next to the elasticity, perhaps the phenomena of the viscosity of gases are best adapted to investigation.¹

Thesis supervisor: Stefan Marks

Title: Associate Professor

Thesis supervisor: Jairo Gutierrez

Title: Professor

¹Text from Holman (1876): doi:[10.2307/25138434](https://doi.org/10.2307/25138434).

Acknowledgments

Write your acknowledgments here.

Biographical Sketch

Silas Whitcomb Holman was born in Harvard, Massachusetts on January 20, 1856. He received his S.B. degree in Physics from MIT in 1876, and then joined the MIT Department of Physics as an Assistant. He became Instructor in Physics in 1880, Assistant Professor in 1882, Associate Professor in 1885, and Full Professor in 1893. Throughout this period, he struggled with increasingly severe rheumatoid arthritis. At length, he was defeated, becoming Professor Emeritus in 1897 and dying on April 1, 1900.

Holman's light burned brilliantly before his tragic and untimely death. He published extensively in thermal physics, and authored textbooks on precision measurement, fundamental mechanics, and other subjects. He established the original Heat Measurements Laboratory. Holman was a much admired teacher among both his students and his colleagues. The reports of his department and of the Institute itself refer to him frequently in the 1880's and 1890's, in tones that gradually shift from the greatest respect to the deepest sympathy.

Holman was a student of Professor Edward C. Pickering, then head of the Physics department. Holman himself became second in command of Physics, under Professor Charles R. Cross, some years later. Among Holman's students, several went on to distinguish themselves, including: the astronomer George E. Hale ('90) who organized the Yerkes and Mt. Wilson observatories and who designed the 200 inch telescope on Mt. Palomar; Charles G. Abbot ('94), also an astrophysicist and later Secretary of the Smithsonian Institution; and George K. Burgess ('96), later Director of the Bureau of Standards.

Contents

List of Figures

List of Tables

List of Acronyms

VR Virtual Reality

AR Augmented Reality

TEL Technology Enhanced Learning

MR Mixed Reality

CSCW Computer Software for Cooperative Work

Chapter 1

Introduction

In the Netflix animated series *Cyberpunk: Edgerunners* (Imaishi et al., 2022), the main character attends courses at a prestigious academy that uses advanced technology to boost education and performance. The students gather in a room with **VR!** (**VR!**) headsets and connect to the lesson, which is taught by a holographic (and probably powered by artificial intelligence) tutor. The episode does not show the lesson itself (the entire system fails catastrophically when attacked by malware), but the brief scene highlights the vision many people have of a futuristic classroom, one guided by advanced technology and automatization.

Forsler et al. (2024) propose the term “post-digital classroom” to identify the characteristics and trends in education present in a world beyond the adoption of digital technologies. The purpose of this classroom is not anymore to introduce new technologies to students, but to implement them as an essential factor of the learning process. The post-digital classroom is interconnected, social and global. In this scenario, learning goes beyond the physical classroom because it is based on creating relationships between concepts and developing valuable skills rather than acquiring knowledge. It is also a classroom where information is detached from the physical learning institute, and access to facts and sources is, ideally, immediate, ubiquitous and democratic.

The classroom has also become hybrid. There is little distinction between network-based lessons and the face-to-face spaces associated with schools and universities (Goodyear et al., 2004). Hybrid and blended approaches blur the distinctions between learning in an online setting and learning in a classroom. In both instances, information is at hand in the network, students can easily connect to peers and mentors, distance and time constraints are less meaningful and knowledge is presented in a multimedia format that needs a technological backbone to be created and shared (Rivoltella, 2008; Shank, 2005).

This thesis explores the use of technology in such post-digital classrooms, specifically, **AR!** (**AR!**) within the context of mobile computing. Moreover, this research will analyse the creation of effective learning spaces, those places where students can build, personalise and control all the activities they engage with (Gourlay & Oliver, 2016). Learning spaces are interesting tools at the disposal of students because these are spaces that can exist outside the norm of a typical classroom, they are more fluid, less structured and can easily connect to other similar spaces (Nørgård & Hilli, 2022). The proposal of this research is that **AR!** can be used to boost the way students build and interact with their own learning spaces and can also offer educators another option to be integrated in the construction of a lesson that

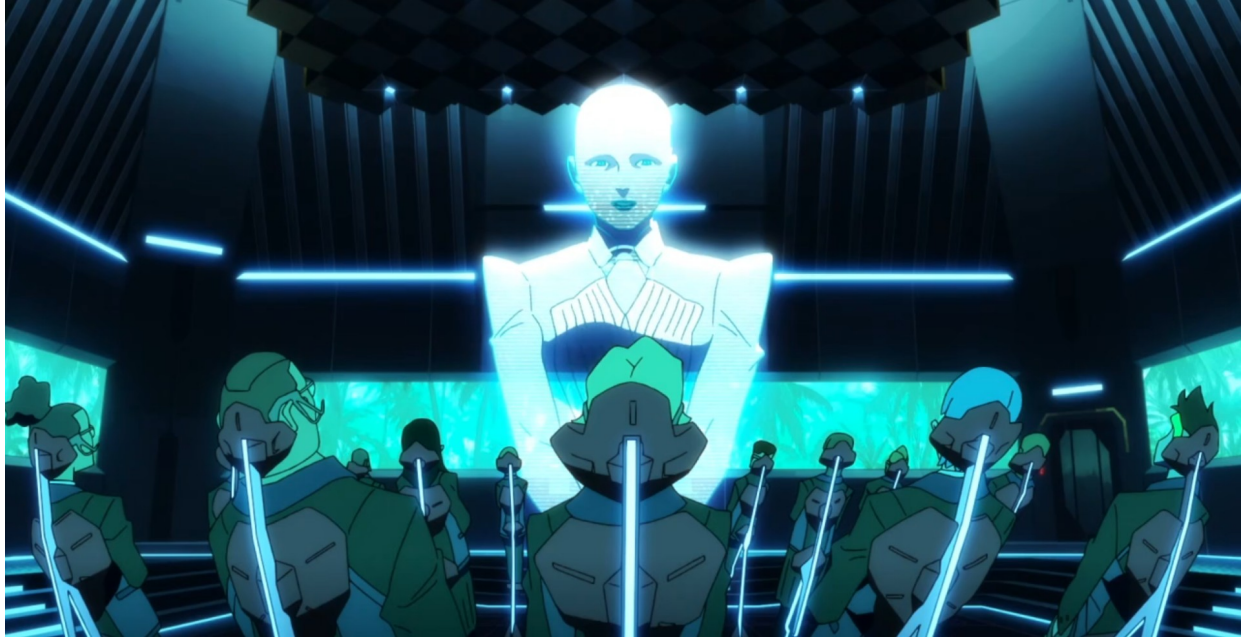


Figure 1.1: Virtual Classroom in Cyberpunk: Edgerunners

capitalises the characteristics of the aforementioned post-digital classrooms.

Within the field of **TEL!** (**TEL!**), **AR!** and related technologies were chosen as a focus of interest because of their intrinsic relationship with space. The initial example scene of Cyberpunk: Edgerunners serves to illustrate the collective idea of what could be achieved in the future with technologies like **VR!** and **AR!**: the possibility of creating an entire new space, suited for the needs of the classroom or the students. It is a space that is shared, in which information and interactions flow between users and that can be controlled at will by students and teachers. In reality, the technology is not quite there yet, especially in terms of fidelity and immersion (Checa & Bustillo, 2020; Hamad & Jia, 2022). What can be explored right now is the creation of interesting synthetic environments that facilitate or allow experiences that are not possible in the common classroom. This process is limited more by the creativity and expertise of the developer rather than by the technology, and therefore can be explored easier and in more detail.

AR!, compared to **VR!**, has a more direct relationship with the *real*, physical space and with the context of the user. Instead of an isolated immersion, **AR!** creates a connection with the environment based on image detection, data and visualizations. This natural expression of the technology is where the value of **AR!** can be explored in relation to the creation of innovative learning spaces. The technology not only has interesting capabilities for visualization and interaction, it also allows an easier implementation of shared, social components that can be used to build activities based on cooperative simulations, sharing knowledge and the creation of communities, all crucial elements for the construction of effective learning spaces (Bligh & Crook, 2017).

The research is positioned then in the conjunction of these three fields: Learning spaces, **TEL!** and **AR!**. The guiding goal is to understand how students are using technology to learn, and how **AR!** can be used to boost that process. **AR!** shines at creating immersive

experiences that can be shared with others and that consider the physical context of the user. I want to use this collaborative space as an educational tool that takes advantage of the inherent social aspects of learning, building and sharing knowledge.

No modern classroom is free of technology, the simple act of taking notes on a paper already implies a set of technological developments needed to put knowledge into written form. Networking and information technologies are just another set of tools at the disposition of any student. A personal learning space can incorporate different software tools, many of which are now ubiquitous and almost indispensable in any learning activity: text processors, multimedia editing tools, web browsers and document readers, plus any specialised tool that the learning subject requires. But that learning space is also built around more elements, some of them equally indispensable for some students. People need a space to work, access to different amenities, access to textbooks, information and teachers. Some may need a quiet space to concentrate, or, on the contrary, may need access to music, to snacks or to a separate place to periodically take a break. The construction of a learning space involves many different elements, some more technological than others, and its effectiveness is based more on the integration of those elements than anything else (Lu & Churchill, 2013; Reinius et al., 2021). This research aims at identifying how to properly integrate **AR!** in the construction of a learning space, which is already composed by a complex web of many other objects, tools and circumstances.

As a researcher. I have explored in the past the way different technologies can be used as part of those learning spaces, either with tools for online learning scenarios or using **VR!** to create a digital space to explore multimedia content (Gil Parga, 2015). This current project can be seen as a way to expand on the work done before, identifying ways to adapt a different technology, to apply lessons learned and to explore new avenues of research. In particular, it is an opportunity to expand from personalized learning environments into social learning spaces, and explore the role of the community in teaching and learning.

It was clear in the work done previously and through some notable literature like Richards (2023) and Long and Ehrmann (2005) that the construction of learning spaces is always mediated by the community around us, even when creating a space as personal as possible. Even nowadays that we live in a post-pandemic world where we seek remote, personalised and secure ways of learning, it has become clear that the social aspect of learning is a crucial element in the equation (Baber, 2021; De Felice et al., 2023). The aim of this research is to delve deeper into the social elements that influence the construction of a learning space, how the students negotiate the process of learning with others and the benefits of collaboration, as well as the challenges it conveys.

This is also a technological enquiry. Having explored **VR!** in the past, it was clear that this family of technologies offer interesting advantages by using complex visuals, new forms of presenting information and more natural interactions suited for the exploration and experimentation with 3D spaces. These technologies also present curious challenges, especially in the usability front. For instance, the public has not yet developed a strong relation or consistent literacy with the technology (Özgen et al., 2021), which creates accessibility problems. And those technologies that leverage common and accessible platforms, like mobile devices, risk carrying with them problems associated with the excessive and pervasive presence of such technologies (Forsler & Guyard, 2025; Petrucco, 2021).

The following introductory chapter will expand this brief exploration of the context and

motivation for the research proposed, outlining the main research goal and the reasoning behind the formulation of the selected research questions. There will be a brief exposition of the methodology followed to build possible answers, the significance of this research and why the answers to the questions proposed can provide important insights about the way we use technology to learn and how we design technological tools for the classroom. It will be followed by a summary of the results obtained and the contributions achieved, as well as the scope and limitations of the work done. The chapter will conclude with a description of the structure of this thesis document.

1.1 Context

AR! technologies are positioned somewhere in the middle of the reality-simulation spectrum proposed by Milgram et al. (1995), a model for display technologies used to show digital or physical data in virtual or synthetic realities. The main characteristic of **AR!** in this spectrum is the incorporation of the real world as part of the experience, making use of different image recognition and composition techniques to identify features in the physical space and overlay them with digital information.

Schmalstieg (2016) adds another layer of analysis to the description of **AR!**-like technologies by using the ubiquitous computing spectrum proposed by Weiser (1991). The model positions different implementations of **AR!** depending on the use of more static or more mobile deployment platforms. In one end of the spectrum are applications that can only be used in singular static terminals, while the other side contemplate platform-independent mobile technologies and ubiquitous systems. Figure ?? shows examples of this distribution as proposed by the authors.

In this bi-directional spectrum we can find mobile **AR!** as a family of technologies that offer an in-between mixture of virtual and physical elements in which location, place and context are additional inputs used for interaction with the synthetic reality. Any technology in the spectrum will offer a variety of characteristics that align with one or more descriptions at a time. **MR!** (**MR!**), for example, are seen as an extension of both **VR!** and **AR!**, offering seamless interaction and connectivity between the real and the simulated world. It is a blurry line that often must be analysed in a case-by-case approach. No matter what label is used to describe a technology, some agreed upon characteristics must be present for it to be categorized as **AR!**, as stated by Azuma (1997):

- A combination of real space and digital information.
- Interaction with the simulation in real time.
- The simulation is done in a 3D space.

With these descriptors it is possible to position the used of the technology in different fields, independent of the labels used to identify it. Based on the themes introduced in the opening statement of the chapter, the goal is to analyse **AR!** as **CSCW!** (**CSCW!**). For this task, a useful analytical tool is the classification of **CSCW!** technologies proposed by Rodden (1991) in terms of the point in time and space where the collaboration is taking

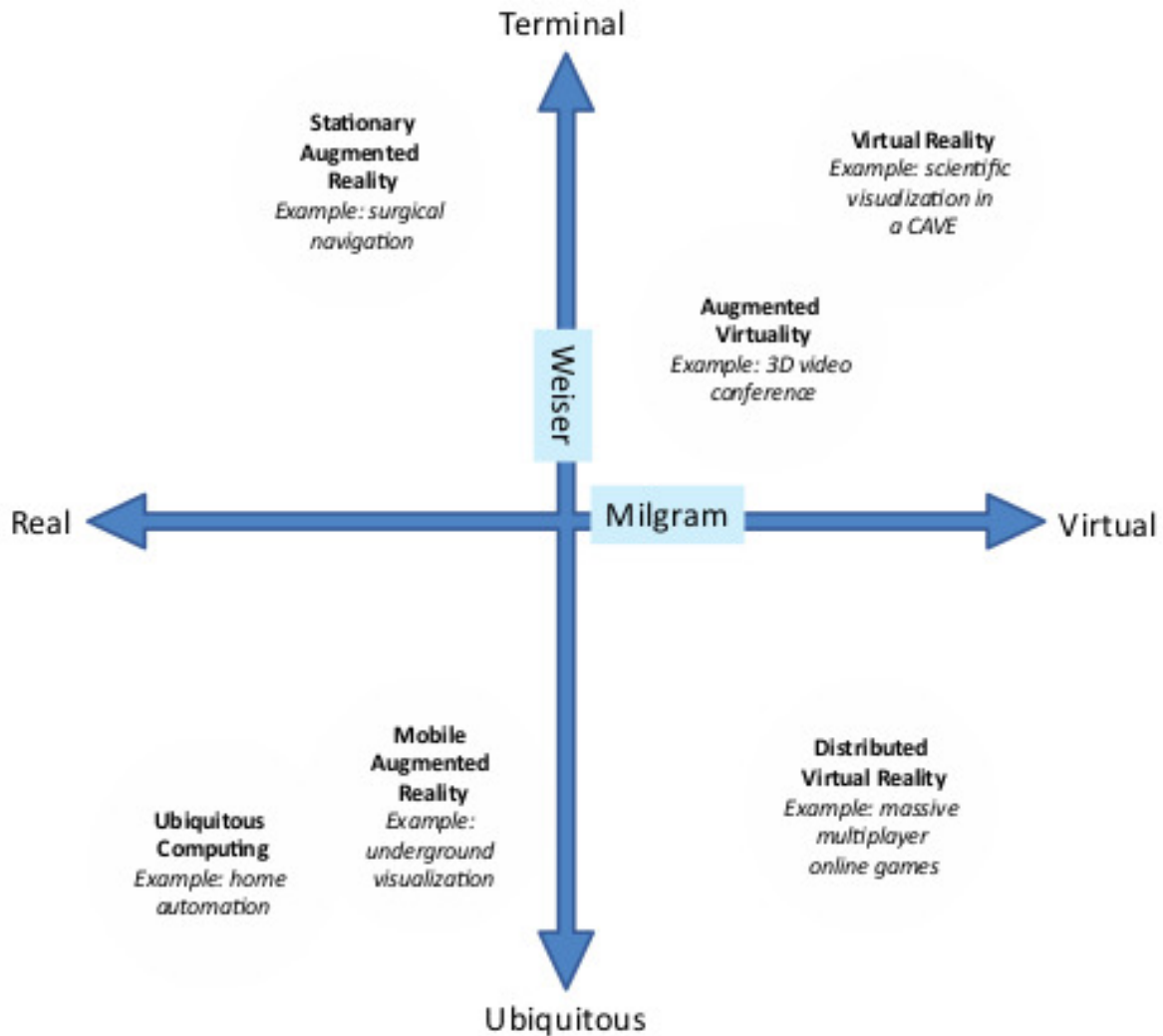


Figure 1.2: AR! in the Milgram-Weiser Spectrum (Schmalstieg, 2016).

place. Figure ?? shows examples of different systems that can be found in each quadrant of the classification, which indicate whether the activity is synchronous or asynchronous, remote or co-located.

Complementarily, Kiyokawa et al. (2002) offer two important elements present in a collaborative activity: communication and tasks spaces. In the communication space, people exchange information and can sustain formal and informal conversations. In the task space, the actual cooperative work is done. Those spaces can be unified or separated, and they can be physical, virtual or in a mixed state. Independent of the nature of each individual space, both need to be present for collaboration to succeed.

Based on these classification models, it is possible to define the main field of interest of this research as the use of **AR!** in a mixed task space for face-to-face collaborative learning, which implies synchronous activities in a co-located space. In this setting it should be possible to use the capabilities of **AR!** for visualization and interaction with the physical space to

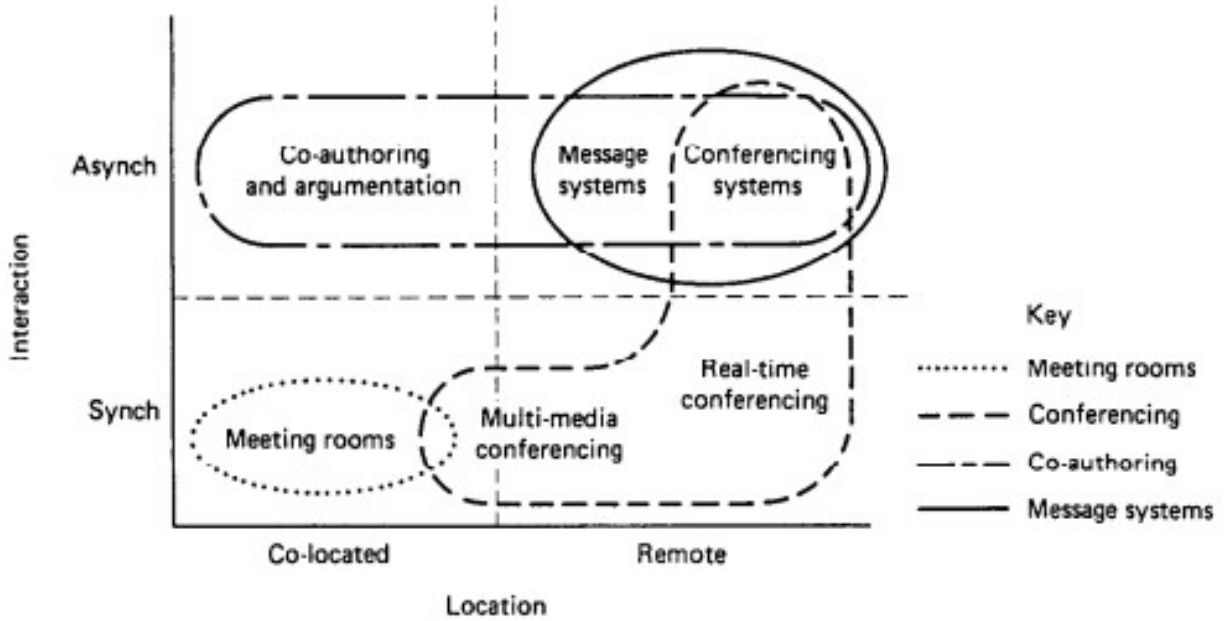


Figure 1.3: Classification of **CSCW!** Systems by Rodden (1991).

augment the collaboration process in terms of:

- Activities related to the communication space, the flow of information and the creation of knowledge in the group.
- The use of the physical space as an asset in the collaborative process.
- Management of the collaborative process itself and the structures that can be created to facilitate effective teamwork and cooperation.
- Guidelines to acquire and refine skills related to teamwork and collaboration.

The main proposal is that the unique characteristics that **AR!** provides in the field of **MR!** are well suited to experiences in co-located collaboration, and that a learning design that is using a collaborative approach can benefit from an **AR!** tool mediating in the communication and tasks spaces of the activity.

The interest in collaborative learning derives from the analysis of the use of **AR!** in education and the tendencies identified in the available literature. It was possible to note that **AR!** is often used as a tool to explore concepts and practices in hands-on approaches that apply previous knowledge in different situations and contexts. Unfortunately, the technology is seldom used in cooperative scenarios, despite the tendency to use mobile platforms for deployment (Gil Parga et al., 2023).

Collaborative learning is a subject rich in discussion. The conversation can focus on the benefits of groupwork in the classroom like promoting different points of view, facilitating complex understanding of a topic and creating the space for an integral development of the student (Smith et al., 1992; Van Wyk & Haffejee, 2017). On the other hand, the challenges associated with collaborative work are also amply explored (Baloche & Brody, 2017; Buchs

et al., 2017; Popov et al., 2012), and there is plenty discussion around issues related to classroom management, inclusivity, effective assessment and good practices.

Many studies have identified and compiled the advantages of using **AR!** and **MR!** as learning tools. For instance, Penn and Ramnarain (2023) offer an analysis of the educational foundations used to design learning interventions. They conclude that most projects converged into constructivist foundations and embodiment theory, opting to create spaces for students to experience and explore the learning subject, with an emphasis on immersion, practice and hands-on interactions. Despite these commonalities, the authors also remarked that most reports do not mention explicitly or do not rationalize the theoretical background of the intervention. They also note that a lot of researchers still hold the idea that introducing a new technology in the classroom is enough to reap its benefits, without caring to build or follow a proper integration plan.

This issue acquires even more relevance now that the mediation of different technologies in the collaboration process has grown due to the Covid-19 pandemic, which forced a quick and often poorly planned implementation of remote learning (Drueke et al., 2021; Nuere et al., 2021). Over time, this prompted the re-design and re-thinking of how to use technology to support the sudden disruption in learning methodologies and approaches (Adi Badiozaman et al., 2020; Fischer et al., 2020). The forced isolation also highlighted the importance of social spaces in the constructions of learning environments, and the need to provide those social interactions even in online spaces (McInerney & Roberts, 2004; Walas-Trębacz et al., 2025).

The post-pandemic classroom is dealing now with the need to balance the advantages of online spaces with the problems associated with socialization solely mediated by technology, which tends to be asynchronous and impersonal. There is value in the physical space for the learning process (Paechter & Maier, 2010; Petchamé et al., 2021; Thai et al., 2020), and a technology that is able to connect the physical and the digital is worthy of a deeper exploration to identify proper approaches and successful use cases.

The idea is not to oversimplify the presence of digital technologies in the classroom, nor follow a misguided tendency of selling all the benefits of a technology without acknowledging its problems. If the proposal is to identify the effects of a collaborative learning design mediated by **AR!**, it is important to also identify all the problems that derive from this approach, either because of the issues that naturally come from teamworking scenarios in the classroom or the negative behaviours that the excessive use of digital technologies promotes, especially in the specific environment of mobile platforms.

With that context just detailed, it is possible to summarize the goal of this research as: The construction of an **AR!** tool that serves as aid in a collaborative learning setting by helping students to communicate and develop teamworking skills. This can be achieved by means of interactivity, visualization, immersion and augmentation of the physical space.

1.2 Research Structure

The goal of the research can be justified by an abundance of literature showing the elements of **AR!** used in the past to offer distinctive value for learning, and the open opportunities to explore and understand better the development of multi-user **AR!** experiences for collaboration.

The nature of the proposal implies that the research has two main fronts:

- In the *technological* front, the goal is to build a software solution that considers the development challenges of an **AR!** tool in a multi-user and multi-platform environment, and the specific requirements derived from the learning goals proposed in the educational front.
- In the *educational* front, the goal is to identify, describe and develop a case study based on collaborative learning, and to analyse the behaviours that students develop when learning in a collaborative environment mediated by technology, comparing the use of **AR!** with other tools or approaches.

1.2.1 Research Questions

The guiding research questions of this project are aimed at obtaining information of what constitutes an effective collaborative learning design using **AR!**. The answers can be stated in terms of learning goals, interaction design and integration plans with the classroom. The questions also seek information about how **AR!** is creating an effect in the behaviour of the students while using the tool for learning. These behaviours can be related to communication, socialization, planning and engagement with the learning material. The main research question will be stated as follows:

MQ: How does the behaviour of a group of students changes when using an **AR!** tool designed to aid in the achievement of a learning goal framed in a collaborative learning scenario?

To guide the implementation of the tool through a strong design, a lot of information can be extracted from current literature and from previous projects using similar approaches. Information gathered from real and incremental usage off the system also offers valuable insights in terms of usability, viability and end-user perception. With this in mind, an iterative development process is proposed to facilitate the exploration of ideas regarding interaction design and a proper fit to the learning scenario where the tool is going to be used.

The implementation of the tool alone is insufficient to answer the main question. For this purpose, a case study research methodology is proposed with the objective of identifying and describing a strong context in which collaboration is used for learning and in which the implementation of the **AR!** tool can be contrasted and compared to other approaches. In this context it is also easier to observe more natural behaviours from the students and the observations will show results closer to how the tool could be used in a real deployment.

It is very important to carefully select a case study that not only offers unique circumstances valuable enough to be analysed in detail but also has the proper learning context to explore the themes of interest of this research project. Although the specific learning field of a prospective case study is not that important, it needs certain characteristics, independently if it is developed through a single activity, a section of a course or through the whole course. The case study must incorporate all the following elements:

- A clear learning goal, either as a whole or as a step to achieve a more complex goal. Assessment activities alone are not enough because the research objectives require the observation of collaborative learning, not the development of a single teamwork activity.
- A collaborative approach to the learning activity. Students must be expected to work in groups to achieve the task, or at least, to have access to a space for collaboration in the development of the activity, even if it is individual in nature.

Ideally, the case study also contemplates the process of learning to collaborate, either as its own learning goal or willing to implement such an objective as a secondary goal. This is not essential, but the current proposal borrows a lot of elements from learning designs about developing collaborative skills. If the selected case study already seeks learning objectives in this venue, then the proposed intervention could offer more value.

The selected case study offers an excellent opportunity, not only to experiment with the proposed thematic of collaboration and technology mediated learning, but because it is also a unique and interesting context in terms of learning design and the opportunities for coordinated teamwork, project development and innovative use of technology.

The Industry Project course of the Built Environments program is a complex scenario that provides students with opportunities to develop industry-level skills through a project-based approach. It presents the opportunity to intervene in the whole course and to use the software solution proposed across different stages of development of the course and different learning objectives, an ideal scenario for varied observations and a rich space for experimentation and discussion.

A set of secondary question can be described that will help to focus the data gathering stages of the research, especially those related to the observations and activities in the context of the case study:

SQ1: What differences in behaviour can be observed between students collaborating using the **AR!** tool and those that are not?

SQ2: What differences the students report in their perception of the activities done in the course between those using the **AR!** tool and those who are not?

SQ3: How does the collaboration and communication process changes between students using the **AR!** tool and those that are not?

SQ4: What differences can be observed in the end results of the course between all the different collaborative approaches observed, especially of those that incorporate the **AR!** tool?

The focus of the observation process should be in identifying behaviours and processes rather than only differences in the end results, that is, the final assessment of the course project. Relying solely on possible improvements to grades or assessments can be a valid but flawed tool to understand the effects of a technology in the classroom (**katz_2017; dalziel_1998**). Additionally, the selected case study proposes a complex course to analyse, where several and diverse factors influence the work of the students. Taking this into consideration is why a

positive or negative result does not tell the full story and why is more important to observe the whole learning process, from planning to development, documenting the behaviours developed to fulfil tasks and reach goals. The objective is to identify if the **AR!** tool proposed is an influence, either positive or negative, and if it provides a more meaningful impact to the learning process as a whole and not only to the end assessments.

Assuming there are actual substantial differences between the students using the **AR!** tool and those that are not, then it is also important to ask:

SQ5: What elements of a technological intervention using **AR!** create the most impact in changing the behaviours, perceptions or results of the students?

SQ6: What elements of a multi-user **AR!** tool creates the most impact in changing the behaviours, perceptions or results of students collaborating and communicating in a learning scenario?

These last questions are meant to provide an evaluation to the decisions taken in the development of the **AR!** tool, especially the multi-user interactions, and how the tool was integrated to the learning design of the course. It also offers the possibility to highlight elements of the tool that could be polished or reframed in future work.

1.2.2 Scope and Limitations

Using a case study methodology, it is possible to approach the research questions through a qualitative lens, which is better suited for understanding the information gathered about behaviours, relationships developed between students and their subjective perceptions of the course, the activities developed, and the results obtained. All those elements would be hard or contra productive to try and quantify. In this light, it is important to acknowledge the limitations in scope and reach derived by the selected methodologies.

In relation to the software development aspect, it is important to establish the scope of the product as a prototype that explores the ideas of collaboration and multi-user **AR!**. A thorough analysis of the design and architecture of the software was conducted to describe different possible approaches and solutions suited to the context and the needs of the tool, keeping an analytical mindset for all the ramifications and consequences derived from the implementation of a particular idea or design.

In terms of the concrete implementation of the software, the work was undertaken by a sole developer and limited in time and resources related to access to the case study. Most of the design and architectural proposals were tested via mock-ups or wireframe prototypes (coleman_2017; sutipitakwong_2020) in pseudo-structured test scenarios, aiming primarily for quick feedback and impressions rather than structured data. This allowed constant iteration and exploration of different ideas. The final implementation of the software prioritized showing the most critical use cases and the most relevant interactions with the tool, most of them tailored for specific observation sessions.

This approach had an important influence in the structure of the observation process and the nature of the data gathered, fomenting a more guided use of the tool and a more structured response from the participants rather than the free observation initially intended.

This was an acceptable concession made to facilitate the development process and to give the participants early and consistent access to the tool, albeit in a more incomplete state. This also minimized the disruption to academic work while maximizing the opportunities for meaningful observation in the available time.

There is also an inherent problem in the development of a case study in relation to the applicability of the results, generalization and subjectivity. The Industry Project course was chosen because it fulfilled all the characteristics needed for the research, but convenience aside, the course also provides a unique opportunity to analyse a collaborative learning design in action, with scenarios that could be extrapolated to other similar contexts, but also a unique take on the design of project-based learning and assessment, as well as a complex context for implementation and challenges that had to be tackled, all elements worth of a detailed analysis.

Nonetheless, it is important to identify the limitations of using one sole case study, which can be summarized as issues with the transferability and applicability of the data gathered. Other factors that impact heavily the nature of the data are the overall seniority of the students, which were all close to finish their studies and the lack of a concrete study plan of the course, which focused more on the development of the project as its main teaching instrument, offering just a relatively vague and general set of learning goals.

Chapter 3 will describe in more detail the scope and limitations of the case study, as well as the measures taken to minimize or acknowledge the flaws and restrictions of the chosen framework. For now, these issues can be summarized the main limitations related to focusing the software development into a gradual an iterative progression of a prototype and the validity of the case study in terms of the characteristics needed for the research and the unique value it offers for analysis and discussion.

1.3 Significance and Contributions

The technological environment in which we found ourselves right now evolves rapidly and at large steps. Despite critics of stagnation, the classroom evolves and adapts in response and constantly tries to understand how the available technologies can improve the way we learn (**watters_2021**). It is not always a positive relationship, it is often marred by overselling products and forcing a technological implementation without a plan and a proper consideration of the context and needs of the classroom (**odunaike_2013**; **cochrane_2012**). Currently, it is also a field that experiences a lot of social scrutiny as we react to the idea that we have become somewhat addicted to digital content and to the devices used to consume it (**liu_2022**; **gui_2021**). In a context in which we are banning the use of smartphones in the classrooms and obtaining somewhat positive results because of it (Rahali et al., 2024), why, then, rely on a proposal that fundamentally uses that technology?

As **murray_2014**<empty citation> stablish, exposure to technology does not equal comprehension. In an age in which almost all aspects of life are mediate by digital activities, it is important to create a positive, critical and informed relationship with the technology that we have to use in our daily lives, and that relationship does not develops naturally (**selwyn_2009**; **margaryan_2011**). This is a task that can and should be executed in the classroom at different levels of schooling, in different contexts and at appropriate age and

field depths. The classroom is the ideal scenario to create and nourish this relationship in a structured and positive way, yet this is a factor that is seen as lacking in education institutes, especially in middle and high school (**burton_2015; jeffrey_2011**).

1.4 Document Structure

References

- Adi Badiozaman, I. F., Leong, H. J., & Wong, W. (2020). Embracing educational disruption: A case study in making the shift to a remote learning environment. *Journal of Applied Research in Higher Education*, 14(1), 1–15.
<https://doi.org/10.1108/JARHE-08-2020-0256>
- Azuma, R. T. (1997). A Survey of Augmented Reality. *Presence: Teleoperators and Virtual Environments*, 6(4), 355–385. <https://doi.org/10.1162/pres.1997.6.4.355>
- Baber, H. (2021). Social interaction and effectiveness of the online learning – A moderating role of maintaining social distance during the pandemic COVID-19. *Asian Education and Development Studies*, 11(1), 159–171.
<https://doi.org/10.1108/AEDS-09-2020-0209>
- Baloche, L., & Brody, C. M. (2017). Cooperative learning: Exploring challenges, crafting innovations. *Journal of Education for Teaching*, 43(3), 274–283.
<https://doi.org/10.1080/02607476.2017.1319513>
- Bligh, B., & Crook, C. (2017). Learning Spaces. In E. Duval, M. Sharples, & R. Sutherland (Eds.), *Technology Enhanced Learning: Research Themes* (pp. 69–87). Springer International Publishing. https://doi.org/10.1007/978-3-319-02600-8_7
- Buchs, C., Filippou, D., Pulfrey, C., & Volpé, Y. (2017). Challenges for cooperative learning implementation: Reports from elementary school teachers. *Journal of Education for Teaching*, 43(3), 296–306. <https://doi.org/10.1080/02607476.2017.1321673>
- Checa, D., & Bustillo, A. (2020). Advantages and limits of virtual reality in learning processes: Briviesca in the fifteenth century. *Virtual Reality*, 24(1), 151–161.
<https://doi.org/10.1007/s10055-019-00389-7>
- De Felice, S., Hamilton, A. F. D. C., Ponari, M., & Vigliocco, G. (2023). Learning from others is good, with others is better: The role of social interaction in human acquisition of new knowledge. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 378(1870), 20210357. <https://doi.org/10.1098/rstb.2021.0357>
- Drueke, B., Mainz, V., Lemos, M., Wirtz, M. A., & Boecker, M. (2021). An Evaluation of Forced Distance Learning and Teaching Under Pandemic Conditions Using the Technology Acceptance Model. *Frontiers in Psychology*, 12.
<https://doi.org/10.3389/fpsyg.2021.701347>
- Fischer, G., Lundin, J., & Lindberg, J. O. (2020). Rethinking and reinventing learning, education and collaboration in the digital age—from creating technologies to transforming cultures. *The International Journal of Information and Learning Technology*, 37(5), 241–252. <https://doi.org/10.1108/IJILT-04-2020-0051>

- Forsler, I., Bardone, E., & Forsman, M. (2024). The Future Postdigital Classroom. *Postdigital Science and Education*. <https://doi.org/10.1007/s42438-024-00488-y>
- Forsler, I., & Guyard, C. (2025). Screens, teens and their brains. Discourses about digital media, learning and cognitive development in popular science neuroeducation. *Learning, Media and Technology*, 50(2), 191–204. <https://doi.org/10.1080/17439884.2023.2230893>
- Gil Parga, S. (2015). *Ambientes virtuales de aprendizaje aplicados* [Doctoral dissertation, Universidad de los Andes]. Retrieved July 7, 2025, from <https://repositorio.uniandes.edu.co/entities/publication/c5cf341b-1dfe-43bd-ac7c-61294516c8c2>
- Gil Parga, S., Singh, U., Gutierrez, J., & Marks, S. (2023). Pedagogical design in education using augmented reality: A systematic review. *Interactive Learning Environments*, 1–18. <https://doi.org/10.1080/10494820.2023.2195445>
- Goodyear, P., Banks, S., Hodgson, V., & McConnell, D. (2004). Research on networked learning: An overview. In P. Dillenbourg, M. Baker, C. Bereiter, Y. Engeström, G. Fischer, H. Ulrich Hoppe, T. Koschmann, N. Miyake, C. O'Malley, R. Pea, C. Pontecorovo, J. Roschelle, D. Suthers, P. Goodyear, S. Banks, V. Hodgson, & D. McConnell (Eds.), *Advances in Research on Networked Learning* (pp. 1–9). Springer Netherlands. https://doi.org/10.1007/1-4020-7909-5_1
- Gourlay, L., & Oliver, M. (2016). Students' Physical and Digital Sites of Study: Making, Marking, and Breaking Boundaries. In *Place-Based Spaces for Networked Learning* (pp. 73–86). Routledge.
- Hamad, A., & Jia, B. (2022). How Virtual Reality Technology Has Changed Our Lives: An Overview of the Current and Potential Applications and Limitations. *International Journal of Environmental Research and Public Health*, 19(18). <https://doi.org/10.3390/ijerph191811278>
- Imaishi, H., Kaneko, Y., Pondsmith, M., Sztybor, B., & Ôtsuka, M. (2022, September). Cyberpunk: Edgerunners (episode 1).
- Kiyokawa, K., Billinghurst, M., Hayes, S., Gupta, A., Sannohe, Y., & Kato, H. (2002). Communication behaviors of co-located users in collaborative AR interfaces. *Proceedings. International Symposium on Mixed and Augmented Reality*, 139–148. <https://doi.org/10.1109/ISMAR.2002.1115083>
- Long, P. D., & Ehrmann, S. C. (2005). The future of the learning space: Breaking out of the box. *EDUCAUSE review*, 40(4), 42–58.
- Lu, J., & Churchill, D. (2013). Creating personal learning environments to enhance learning engagement. *2013 IEEE 63rd Annual Conference International Council for Education Media (ICEM)*, 1–8. <https://doi.org/10.1109/CICEM.2013.6820194>
- McInnerney, J. M., & Roberts, T. S. (2004). Online learning: Social interaction and the creation of a sense of community. *Journal of Educational Technology & Society*, 7(3), 73–81.
- Milgram, P., Takemura, H., Utsumi, A., & Kishino, F. (1995). Augmented reality: A class of displays on the reality-virtuality continuum. *Telemanipulator and Telepresence Technologies*, 2351, 282–292. <https://doi.org/10.1117/12.197321>
- Nørgård, R. T., & Hilli, C. (2022). Hyper-Hybrid Learning Spaces in Higher Education. In *Hybrid Learning Spaces* (pp. 25–41).

- Nuere, S., Suz, A. A., & Álvarez, L. d. M. (2021). Reflections on the adaptation to the COVID-19 pandemic in higher education. In *Remote Learning in Times of Pandemic*. Routledge.
- Özgen, D. S., Afacan, Y., & Sürer, E. (2021). Usability of virtual reality for basic design education: A comparative study with paper-based design. *International Journal of Technology and Design Education*, 31(2), 357–377.
<https://doi.org/10.1007/s10798-019-09554-0>
- Paechter, M., & Maier, B. (2010). Online or face-to-face? Students' experiences and preferences in e-learning. *The Internet and Higher Education*, 13(4), 292–297.
<https://doi.org/10.1016/j.iheduc.2010.09.004>
- Penn, M., & Ramnarain, U. (2023). A Systematic Review of Pedagogy Related to Mixed Reality in K-12 Education. *Mixed Reality for Education*, 85–108.
- Petchamé, J., Iriondo, I., Villegas, E., Riu, D., & Fonseca, D. (2021). Comparing Face-to-Face, Emergency Remote Teaching and Smart Classroom: A Qualitative Exploratory Research Based on Students' Experience during the COVID-19 Pandemic. *Sustainability*, 13(12), 6625. <https://doi.org/10.3390/su13126625>
- Petrucchio, C. (2021). Smartphones in and Outside Classroom. In *Teaching and Mobile Learning*. CRC Press.
- Popov, V., Brinkman, D., Biemans, H. J., Mulder, M., Kuznetsov, A., & Noroozi, O. (2012). Multicultural student group work in higher education. *International Journal of Intercultural Relations*, 36(2), 302–317. <https://doi.org/10.1016/j.ijintrel.2011.09.004>
- Reinius, H., Korhonen, T., & Hakkarainen, K. (2021). The design of learning spaces matters: Perceived impact of the deskless school on learning and teaching. *Learning Environments Research*, 24(3), 339–354. <https://doi.org/10.1007/s10984-020-09345-8>
- Richards, T. (2023). Reimagining the classroom : Creating new learning spaces and connecting with the world /.
- Rivoltella, P. C. (2008). Introducing Multimedia in the Classroom. In *Empowerment Through Media Education: An Intercultural Dialogue* (pp. 201–210). The International Clearinghouse on Children, Youth; Media.
- Rodden, T. (1991). A survey of CSCW systems. *Interacting with Computers*, 3(3), 319–353.
[https://doi.org/10.1016/0953-5438\(91\)90020-3](https://doi.org/10.1016/0953-5438(91)90020-3)
- Schmalstieg, D. (2016). *Augmented reality : Principles and practice* /. Addison-Wesley,
- Shank, P. (2005). The Value of Multimedia in Learning. *ADOBE DESIGN CENTER*, (501), 2–12.
- Smith, K. A., Johnson, D. W., & Johnson, R. T. (1992). Cooperative Learning and Positive Change in Higher Education. In *Collaborative Learning: A Sourcebook for Higher Education* (pp. 46–56). National Center on Postsecondary Teaching, Learning; Assessment, University Park, PA.
- Thai, N. T. T., De Wever, B., & Valcke, M. (2020). Face-to-face, blended, flipped, or online learning environment? Impact on learning performance and student cognitions. *Journal of Computer Assisted Learning*, 36(3), 397–411.
<https://doi.org/10.1111/jcal.12423>
- Van Wyk, J., & Haffeejee, F. (2017). Benefits of Group Learning As a Collaborative Strategy in a Diverse Higher Education Context. *International Journal of Educational Sciences*, 18(1-3), 158–163. <https://doi.org/10.1080/09751122.2017.1305745>

- Walas-Trębacz, J., Krzyżak, J., Herdan, A., Setyohadi, D. B., Jeyanathan, J. S., & Nair, A. (2025). Social interactions and attitudes toward remote learning vs perceptions of its effectiveness in the view of Education 4.0. *The TQM Journal*, ahead-of-print(ahead-of-print). <https://doi.org/10.1108/TQM-09-2024-0335>
- Weiser, M. (1991). The Computer for the 21st Century. *Scientific American*, 265(3), 94–104. <https://doi.org/10.1038/scientificamerican0991-94>